“Machine Learning-Based Analysis of Educational Data”

Using Regression, Classification & Association Techniques

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Introduction

- This project analyzes school dataset using Machine Learning algorithms
- Focus is on:
  - Predicting student performance
  - Classifying schools
  - Finding relationships between facilities
- Multiple ML algorithms are applied for different tasks

Objective

- **To clean and preprocess school dataset**
- **To apply different ML algorithms**
- **To perform:**
    1. Prediction (scores)
    2. Classification (categories)
    3. Pattern analysis
- **To compare results and insights**

Dataset Overview

- Dataset: School_data.csv
- Contains features like:
  - Teacher-Student Ratio
  - Parent Literacy Rate
  - Internet Availability
  - Infrastructure Facilities

- Used for multiple ML tasks

Data Preprocessing

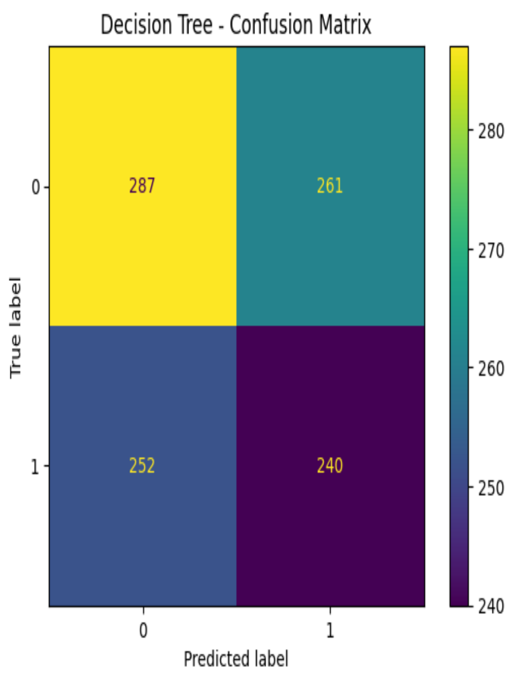
- Missing values handled using mean
- Data cleaned and prepared
- Features selected for different models
- Target variables created:
  Performance_Class
  Area_Class

Algorithms Used

❖This project uses 8 ML algorithms:

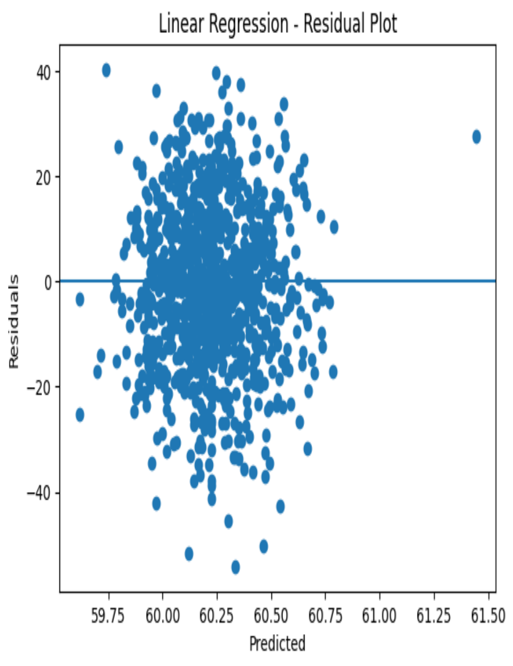
- Decision Tree
- Simple Linear Regression
- Multiple Linear Regression
- Logistic Regression
- Support Vector Machine (SVM)
- K-Nearest Neighbors (KNN)
- Naive Bayes
- Apriori (Association Rule)

Algorithms



Decision Tree

- **Role:**
- Used for classification of student performance
- **Predicts:**
- Performance_Class (Good/Bad)
- **Graph Insight:**
- Tree diagram shows decision paths
- Helps identify which factor affects performance most
- 👉 **Key Point:**
- Clearly explains decision rules in data



Simple Linear Regression

- **Role:**
- Analyzes relationship between one feature and score
- **Predicts:**
- Math Score
- **Graph Insight:**
- Straight line shows positive/negative trend
- Indicates how strongly literacy affects score
- 👉 **Key Point:**
- Shows direct dependency between variables

Algorithms

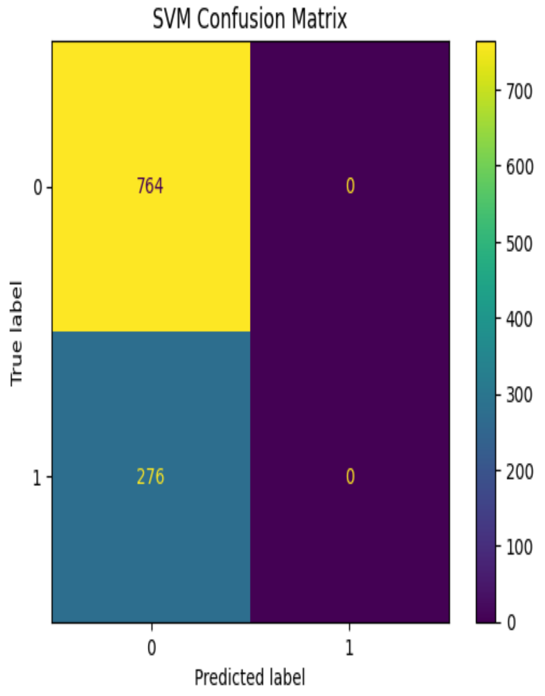
Multiple Linear Regression

- **Role:**
- Analyzes combined effect of multiple factors
- **Predicts:**
- Science Score
- **Graph Insight:**
- Residual plots show prediction accuracy
- Helps understand influence of each feature
- 👉 **Key Point:**
- More realistic analysis using multiple inputs

Logistic Regression

- **Role:**
- Used for binary classification
- **Predicts:**
- Performance_Class (0/1)
- **Graph Insight:**
- S-curve (sigmoid) shows probability
- Helps understand classification threshold
- 👉 **Key Point:**
- Converts data into probability-based decisions

Algorithms



SVM

- **Role:**
- Separates classes using optimal boundary
- **Predicts:**
- Area_Class (Urban/Rural)
- **Graph Insight:**
- Shows margin and separating line
- Maximizes distance between classes
- 👉 **Key Point:**
- Strong classifier for clear separation

KNN

- **Role:**
- Classifies based on nearby data points
- **Predicts:**
- Class label
- **Graph Insight:**
- Cluster of points shows local grouping
- Nearby points influence prediction
- 👉 **Key Point:**
- Relies on similarity between data points

Naive Bayes

- **Role:**
- Uses probability for classification
- **Predicts:**
- Class based on feature likelihood
- **Graph Insight:**
- Probability distribution of features
- Shows feature contribution to prediction
- 👉 **Key Point:**
- Fast and works well with categorical data

Apriori

- **Role:**
- Finds hidden patterns/relationships
- **Analyzes:**
- Facility combinations
- **Graph Insight:**
- Association rules (support/confidence)
- Shows frequent feature combinations
- 👉 **Key Point:**
- Discovers useful patterns, not predictions

Graph Analysis

Regression graphs → show trend between features and scores

Scatter plots → show data distribution & correlation

Classification graphs → show separation between classes

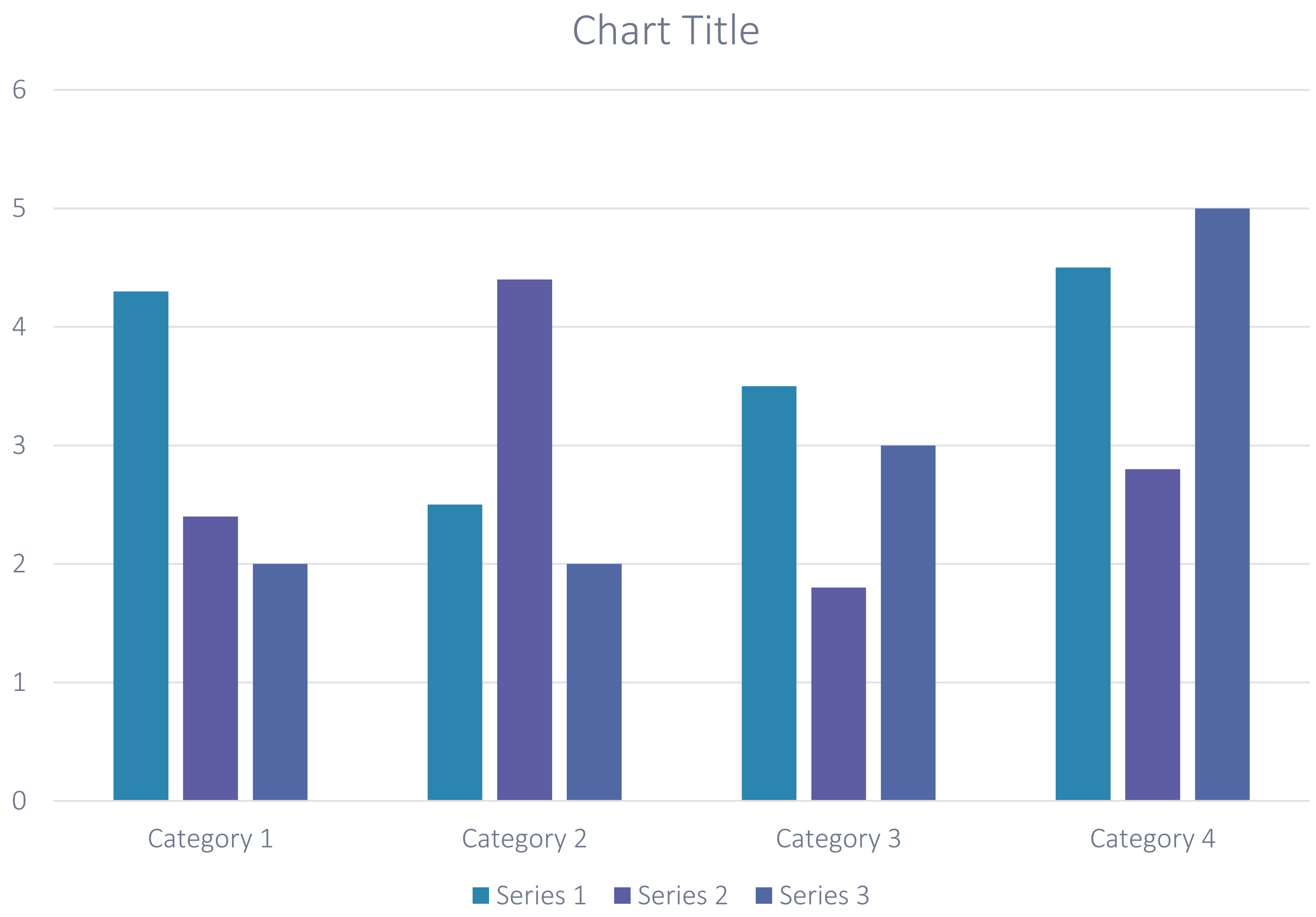
Association rules → show feature relationships

👉 **Final Insight:**

Graphs help understand data patterns and model behavior clearly

Final Results

- Regression models predicted student scores
- Classification models predicted categories
- Association rule found hidden patterns



Conclusion

- Different algorithms are used for different tasks:
- Regression → score prediction
- Classification → category prediction
- Association → pattern discovery
- Data preprocessing improved model performance
- Key Insight:
  👉 Student performance is influenced by literacy, infrastructure, and facilities
- No single model is best for all tasks
- Model selection depends on problem type